m > 1. As it rises, expected unemployment spells with SMI lengthen and SMI costs more. Its moment is the SMI share of the homeless, 25 percent in Gutwinski et al. (2021), since a household out of work longer is likelier to become homeless. μm, the SUD recovery multiplier for SMI. The chain's second direction: a SUD raises the chance of

SMI, and SMI makes a SUD more persistent. As it falls, recovering from a SUD is less likely, and Treatment First has more to overcome. It is set externally at 0.82 from the dual-diagnosis relapse rates in Bradizza et al. (2006). κϕm, the mortality multiplier for SMI. The chain's last link, through which SMI shortens life. As it

rises, more households with SMI die, and Treatment First has more to work on, because less drug use means less SUD, less SMI and fewer deaths. It is set externally at 2.50, the middle of the standardised mortality ratios for SMI in Olfson et al. (2015).

Block 4: Recovery and treatment. Both policies work through this block: Treatment First encourages seeking treatment, and Housing First buys recovery by housing people, since recovery is likelier while housed. The first two parameters set how much recovery there is to buy, the next two how much of it treatment reaches, and the last what treatment costs the government.

¯πr, the baseline SUD recovery rate. This parameter is the annual chance that a housed household re-

covers from a SUD, the level everything else in the block scales. As it rises, disorders are expected to end more quickly on their own, and there is less for either policy to do. It is set externally at 0.025, the annual hazard implied by the ten-year remission rate in Vaillant (1995). μH, SUD recovery when homeless. This parameter sets how much less likely recovery is in a shelter,

and so how much recovery a voucher buys beyond shelter. As it falls, housing buys more recovery and Housing First is more effective in curing a SUD; near one, Housing First has little effect. It targets the effect of housing on recovery in Milby et al. (2005), who find that housing improves recovery chances by 47.3 percent. g, the treatment effect on recovery. The instrument through which Treatment First reaches households

that already have a SUD, which matters most in the short run. As it rises, treatment is likelier to end a SUD, and Treatment First has a larger margin to act on. It is set externally at 2.00, which is how much medication for opioid use disorder raises remission: rates near double those without it (Paul et al., 2024), and randomised trials of agonist medication against placebo find at least as large an effect on abstinence (Fudala et al., 2003). ¯τ u, the utility cost of treatment. Take-up is a choice, and this utility cost is what makes households

decline a treatment that would raise their chance of recovering. As it rises, fewer take treatment up, and Treatment First induces fewer households with a SUD to seek it. It is calibrated internally to treatment take-up: 24.5 percent of adults with a SUD received substance use treatment in the past year in NSDUH 2024. ctreat, the public cost of a year of treatment. This parameter is what the government pays for each

household with a SUD that takes up treatment, under both policies. It ties treatment to the

gov- ernment's budget. Treatment First raises take-up, since entering treatment and stopping is how a household with a SUD keeps its voucher, so part of what the screen saves goes on treatment rather than on vouchers for others. As ctreat rises, Treatment First frees less for vouchers and πV rises less. It is set externally to $9,426 a year, 8.4 percent of average earnings, the cost of a residential treatment episode in Roebuck et al. (2003).

Block 5: Externalities. These two parameters decide what is an externality and how large it is.

ωc, the parent's weight on its child. A parent bears the share ωc of what its choices cost its child, in

childhood and through the productivity penalty. As it rises, the parent counts more of the child's loss, so the budget, earnings and childhood homelessness externalities left for either policy to correct are smaller. It is identified by a counterfactual moment: drug use when the child is given no weight, against drug use at ωc, with every other parameter unchanged.28 The data moment holds the household fixed as well. In NSDUH 2024, an unmarried adult living with one other adult and no child was 17.5 percent more likely to have used illicit drugs in the past year than a lone parent living with one child. The lone parents are reweighted to the other group's age, sex, marital status (never married, divorced or separated, or widowed) and family income; with two members in each household, equal family income is equal income per person.29

28With $\omega c = 0.69$, the weight on children that Nishiyama (2000) estimates and the upper end of the literature, and with b re-matched to drug use, Treatment First still raises a newborn's welfare, by 3.18 percent of lifetime consumption, against 2.57 in the benchmark. Less of that gain is an externality, however: 23 percent of it, against 76 percent in the benchmark (the last column of Table 6), because a parent with this weight counts most of what its choices cost its child, 1.64 percent of lifetime consumption against 0.19. Quantitative models in the dynastic tradition of Barro and Becker (1989) typically use lower weights (Lee and Seshadri, 2019; Abbott et al., 2019).

29This moment counts any past-year use of an illicit drug, a broader measure than the model's, because too few lone parents in the sample have a disorder for the comparison to give a statistically reliable measure. It is therefore used only for how much use responds, in proportion, and not for its level: the model matches it as the proportional rise in its own users.

δd, earnings the child loses because of a parent's SUD. The child inherits not the disorder but a

worse productivity draw, which makes it likelier to be poor, to use, to acquire a SUD and SMI, to be homeless and to die earlier. As it rises, with $\omega c < 1$, the earnings externality grows, which favours whichever policy prevents more drug use: Treatment First by deterring the parent, Housing First by making recovery likelier once housed. It is set externally at 0.24, the earnings loss, at age 32, that Nilsson (2017) measures for children of mothers under 21 exposed in utero to a Swedish alcohol experiment. Section 2 explains why the value is conservative. No study measures the earnings cost of a parent's disorder after birth directly; Nilsson's is the closest causal estimate, and the SUD odds ratio of Block 6 of the validation (Section 5.2) is the indirect check on its size.30

Block 6: Drug market. One parameter makes the drug price endogenous, so that it moves with demand.

ε_s, the price elasticity of drug supply. This parameter sets the size of a general-equilibrium offset that

runs against Treatment First: deterring users lowers demand and the price, and a cheaper drug draws other households in. As it rises, supply is more elastic and the price moves less, so the offset shrinks. I set it to 1.46, the estimate of Mace et al. (2020) for cannabis on Washington State administrative data.31

5.2 Validation

Table 4 compares the model with moments it was not fitted to, in numbered blocks, and each block says why it matters for the experiments.

Block 1: Homelessness. The first row gives each age band's share of the homeless; the youngest band is the target of T0, the other four are free. The model puts 37.9 percent of the homeless at ages 35 to 44, against 27.2 in HUD's PiT count, and 11.5 percent at 55 and over, against 25.9. Of the homeless in the model, 21.4 percent are employed, against 46 in the data, and their income is 6.5 percent of the average, against 19.1 in Meyer et al. (2024). As in the data, most of the homeless are of child-rearing age and poor, so the model places Treatment First's margin, the voucher a household holds or loses, at the ages where it reaches children.

Block 2: Who uses and who has a SUD. Users in the model are somewhat younger than in the data: 67 percent of them are 26 to 34, against 45 in NSDUH 2023, and their mean age is 34.6, against 39.6. Unemployment among households with a SUD is 14.5 percent, against 16.2 in the data. Households with SMI are 2.64 times more likely than others to use drugs, against 2.44, and excess deaths due to SUD account for 14.5 percent of deaths at ages 25 to 44, against 20.0 percent from drug poisonings in the Centers for Disease Control and Prevention (CDC) WONDER database. Both policies reach a child through its parent's decisions, and users are mostly of child-rearing age, in the model as in the data. And since households with SMI use drugs and are homeless at higher rates than the rest, as in the data, the model also captures that Treatment First's screen falls disproportionately on them: they are the likeliest to fail it.

Block 3: Housing and vouchers. In the model, 58.7 percent of housed households spend over thirty percent of their income on rent, against 43.2 percent of renters in the SCF. A voucher holder pays 39.0

30The earnings externality is only 18 percent of the welfare gain from Treatment First (Section 6.3), so the ranking of the two policies does not rest on the penalty's size.

31To the best of my knowledge, there is no direct estimate for an illicit drug, and cannabis is the closest market there is: it remains a Schedule I substance federally, so even a state-licensed producer works without interstate trade, without ordinary banking and under federal prohibition, which is the set of frictions an illicit supplier faces. Mace et al.'s estimate is a short-run one, however: it is the response of Washington's legal retailers in the month after Oregon opened its own retail market in October 2015 (2.03 and 1.76 with

windows of two and three months). It also comes from a market whose producers' licences capped their growing area. The long-run elasticity of an illicit supply, which faces no such cap, is likely higher, and studies of illicit drugs take it to be infinite (Becker et al., 2006).

percent of its income in rent,32 against 31.2 in HUD's Picture of Subsidized Households, and 39.2 percent of voucher holders live mainly on wages, against 28.0; the others are mostly retired or unemployed. Housing takes a large share of poor households' income, so the consumption Treatment First puts at stake when it conditions the voucher is of a similar size in the model and in the data.

Block 4: Income and wealth distributions. The lowest-income fifth of households receives 3.40 percent of income in the model, against 2.71 in the SCF, and holds 0.27 percent of net worth, against -0.23; the bottom half holds 6.3 percent, against 2.2. Both policies act on the households that receive vouchers, the poorest, and the model captures their relative poverty broadly in line with the data.